Product Requirements Document (PRD)

Project: Bubble Gum Dopamine Arcade

1. Primary Users

- **Target Audience:** Casual gamers and general web users seeking quick, high-energy entertainment and instant satisfaction ("dopamine hits") through a gum-themed micro-game experience.
- **Experience Model:** A single, unified user experience across mobile and desktop browsers with no account registration required.

2. Problems to Solve

- **Lack of Instant Interactive Satisfaction:** Users seeking quick, repeatable, and frictionless web micro-experiences lack dedicated interactive tools.
- **Low-Engagement Micro-Gaming:** General casual games often require long setups or complex tutorials rather than immediate gameplay gratification.

3. Capabilities

- **Slider-Based Inflation Mechanic:** A precision-driven slider control allowing players to inflate a virtual bubble, balancing risk and reward to maximize survival time before popping.
- **Economic Progression Loop:** A system that calculates in-game currency earned based on total survival time and active multipliers.
- **Gum Marketplace & Upgrades:** A virtual shop where earned currency can be spent to unlock advanced gum brands with improved performance attributes (e.g., reduced wall shrink speed, higher yield multipliers).
- **High-Dopamine Audio-Visual Feedback:** Integrated sound effects, popping animations, screen shake, and celebratory visual effects triggered by successful inflations and personal records.
- **Local State Persistence:** Automated saving via browser `LocalStorage` to maintain player currency, unlocked inventory, active gum selection, and high scores across sessions.

4. Required Inputs

- **Player Control Inputs:**
    - **Desktop:** Left/Right arrow keys to adjust control position and keep the slider away from the closing walls.

    - **Mobile / Touch:** Direct horizontal touch-and-drag gestures to maintain control position.
- **UI & Menu Inputs:**
    - Touch/Click events for opening the store overlay, selecting/purchasing gum items, equipping inventory, and launching a new inflation attempt.
- **System Inputs & Config Data:**
    - **Persistent Storage Data:** `LocalStorage` JSON object maintaining total currency, unlocked gum IDs, active gum ID, and high scores.
    - **Gum Configuration Data:** Pre-configured item attributes defining price, wall shrink speed modifier, maximum bubble size cap, and currency yield multiplier per gum type.

5. User Flows

- **Flow 1: Core Gameplay Loop**
    1. **Entry:** Player opens the application; saved currency, high score, and equipped gum load automatically from `LocalStorage`.
    2. **Start Attempt:** Player launches a new bubble attempt.
    3. **Survival Mechanic:** Walls begin closing inward. Player continuously uses Left/Right Arrow keys (desktop) or horizontal touch drag (mobile) to keep the slider safe inside the closing boundaries.
    4. **Dynamic Accumulation:** Earnings continuously accumulate second-by-second, scaled by time elapsed and the equipped gum's earning multiplier.
    5. **Pop Event:** When the slider contacts a wall, the bubble pops with audio-visual feedback (pop sound, splash screen/animation, screen shake).
    6. **Summary & Instant Retry:** A brief, punchy round summary overlay displays time survived, round earnings, and updated total currency. The player can immediately press `Spacebar` or tap a single "Play Again" button to restart instantly.
- **Flow 2: Shop & Progression Loop**
    1. **Shop Access:** Player opens the Gum Shop overlay from the main game screen.
    2. **Browse & Compare:** Player inspects locked/unlocked gums, reviewing stats such as price, reduced wall closing speed, and increased currency yield.
    3. **Purchase & Equip:** Player clicks/taps to purchase an item with accumulated currency and equips it.
    4. **Return:** Player closes the shop; new gum stats immediately apply to the next attempt.

6. Constraints

- **Platform & Hardware Scope:** Responsive, browser-based web application optimized for mobile (touch controls) and desktop (arrow key controls).
- **Performance Requirements:** High-framerate rendering (60 FPS) to ensure low input latency and smooth collision detection for closing walls.

- **Session Duration Constraint:** Short, bite-sized rounds designed to cap typical survival times between 10 and 40 seconds via aggressive difficulty and speed scaling.
- **Architecture & Data Storage:** Zero-backend, client-side only architecture. All state persistence relies strictly on browser `LocalStorage` with no user accounts, external databases, or logins required. Offline PWA capabilities are excluded for initial scope simplicity.

7. Non-Functional Requirements & Success Metrics

- **Technical Performance & Stability:**
    - Fast initial load time (under 2 seconds) on standard mobile connections.
    - Consistent 60 FPS gameplay with low-latency collision detection and input response.
    - Flawless local state persistence with zero data loss across browser refreshes or reopens.
- **Dopamine & Engagement Loops:**
    - High sensory polish featuring punchy audio, pop animations, and screen shake on failures or personal records.
    - Frictionless retry mechanism with a round restart delay under 1 second to keep players in the gameplay loop.
- **Usability & Game Balance:**
    - Intuitive controls requiring zero tutorial instructions to learn immediately on mobile or desktop.
    - Balanced economic progression where higher-tier gum purchases provide clear, rewarding stat improvements.